

Operator Discovery Report: pruner_cluster_bottleneck_adaptive

- Operator type: candidate_pruner.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Adaptive candidate-list pruner tuned to keep more 2-opt candidates on bottleneck/two-cluster instances while reducing noise on easy/unstructured ones.
- Novelty classification: trap_expanding_pruner.
- Rediscovery signature: candidate_pruner:20:6:2:3.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Start from a base candidate-list size.
- Expand or shrink the list using instance descriptors such as size, trap score, grid-likeness, clustering, and current stagnation.
- Run the downstream local search only on the resulting pruned neighborhood.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: b8ea304ff7122a53.

Why It Should Help

- The operator adapts neighborhood size to instance structure, which can trade off search breadth and runtime in a reusable way.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.
- Runtime inflation is high at 0.48603.

Transplant Results

- nearest_neighbor_2opt gap delta 0.000609, runtime delta 93.5186, distance-eval delta 3467.428571.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1846.092786, distance-eval delta 1854.285714.
- random_restart_2opt gap delta 0.000974, runtime delta 114.519114, distance-eval delta 7917.142857.
- sparse_three_opt gap delta -0.000253, runtime delta 77.996629, distance-eval delta 5799.857143.
- clustered_local_search gap delta 0.047699, runtime delta 80.014186, distance-eval delta 6013.0.

Instance-Family Results

- heldout_tsplib operator gap 0.235667 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.000224.
- Runtime inflation: 0.48603.
- Distance-evaluation inflation: 0.357171.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.000224.
- constant_pruner mean gap 0.121542 and delta vs full operator 0.0.
- wide_pruner mean gap 0.121542 and delta vs full operator 0.0.